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MATHEMATICS

STANDARD SIX

Annual exam - November 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Six
Year	2024
Exam Board / Issuer	MESP Tanzania
Exam	Annual exam - November
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S6-2024-MESP-Tanzania-November-Annual-exam.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Calculations

1(i). Multiply 94 x 765.

Step 1: Split into place values: $90 \times 765 = 68850$; $4 \times 765 = 3060$.

Step 2: Add the partial products: 71910.

Final Answer: 71910

1(ix). Divide 7320 by 24.

Step 1: $73 \div 24$: write 3; $3 \times 24 = 72$; $73 - 72 = 1$.

Step 2: $12 \div 24$: write 0; $0 \times 24 = 0$; $12 - 0 = 12$.

Step 3: $120 \div 24$: write 5; $5 \times 24 = 120$; $120 - 120 = 0$.

Step 4: $24 \times 305 = 7320$.

Step 5: $7320 \div 24 = 305$.

Final Answer: 305

1(viii). Subtract 4405 - 3897.

Step 1: 1 place: regroup 1 from the next place; $15 - 7 = 8$.

Step 2: 10 place: regroup 1 from the next place; $9 - 9 = 0$.

Step 3: 100 place: regroup 1 from the next place; $13 - 8 = 5$.

Step 4: 1000 place: $3 - 3 = 0$.

Step 5: $4405 - 3897 = 508$; check $508 + 3897 = 4405$.

Final Answer: 508

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Mathematics

1(ii). Divide 170.2 by 74.

Step 1: Multiply both the numerator and denominator by 10: $1702/740$.

Step 2: $740 \times 2 = 1480$, leaving remainder 222; $222/740 = 0.3$.

Step 3: $2 + 0.3 = 2.3$.

Final Answer: 2.3

1(iii). Subtract $5 \frac{2}{3} - 2 \frac{1}{4}$.

Step 1: Convert the mixed numbers to $17/3$ and $9/4$.

Step 2: Use the common denominator 12: $68/12 - 27/12$.

Step 3: $68 - 27 = 41$; $41/12 = 3 \frac{5}{12}$.

Final Answer: $3 \frac{5}{12}$

1(iv). Add $+17 + (-35)$.

Step 1: The signs are opposite: $35 - 17 = 18$. The number with the greater magnitude is negative.

Final Answer: -18

1(v). Divide 0.025 by 2.5.

Step 1: Multiply both numbers by 1000: $25/2500$.

Step 2: $25 / 2500 = 0.01$.

Final Answer: 0.01

1(vi). Subtract $9 - 6.725$.

Step 1: Write 9 as 9.000.

Step 2: Regroup to obtain 8 whole units, 9 tenths, 9 hundredths and 10 thousandths.

Step 3: $10 - 5 = 5$; $9 - 2 = 7$; $9 - 7 = 2$; $8 - 6 = 2$.

Final Answer: 2.275

1(vii). Divide $7\frac{1}{7}$ by $6\frac{1}{4}$.

Step 1: Convert the mixed numbers to $\frac{50}{7}$ and $\frac{25}{4}$.

Step 2: Multiply $\frac{50}{7}$ by $\frac{4}{25}$; cancel the common factor: $\frac{50}{25} = 2$.

Step 3: $2 \times 4 / 7 = \frac{8}{7}$.

Final Answer: $1\frac{1}{7}$

1(x). Subtract $7 - 3\frac{2}{7}$.

Step 1: $7 = \frac{49}{7}$; $3\frac{2}{7} = \frac{23}{7}$.

Step 2: $49 - 23 = 26$; $\frac{26}{7} = 3\frac{5}{7}$.

Final Answer: $3\frac{5}{7}$

2(i). Write two million eight hundred forty-five thousand and twenty-nine in numerals.

Step 1: $2000000 + 845000 + 29 = 2845029$.

Final Answer: 2845029

2(ii). Round 424352 to the nearest thousand.

Step 1: The hundreds digit is 3, which is less than 5. Keep the thousands digit 4 and replace 352 with 000.

Final Answer: 424000

2(iii). Name the place of the underlined 0 in 504251.

Step 1: 0 is the second digit of this six-digit number.

Final Answer: Ten-thousands

2(iv). What is the ten-thousands digit in 7502780?

Step 1: The digits are 7 in millions, 5 in hundred-thousands and 0 in ten-thousands.

Final Answer: 0

2(v). Write CMXCIX in ordinary numerals.

Step 1: $CM = 900$; $XC = 90$; $IX = 9$.

Step 2: $900 + 90 + 9 = 999$.

Final Answer: 999

2(vi). Write 101101 in words.

Step 1: $101101 = 101000 + 101$.

Final Answer: One hundred and one thousand one hundred and one

3(i). Write 37^2 in Roman numerals.

Step 1: $37 \times 37 = 1369$.

Step 2: $1000 = M$; $300 = CCC$; $60 = LX$; $9 = IX$.

Final Answer: MCCCLXIX

3(ii). Find the GCF of 19, 38 and 114.

Step 1: 19 is prime; $38 = 2 \times 19$; $114 = 6 \times 19$.

Final Answer: 19

3(iii). 45% of 800 pupils like mathematics. How many do not like it?

Step 1: $100 - 45 = 55\%$.

Step 2: $800 \times 55 / 100 = 440$.

Final Answer: 440 pupils

4(i). A baby sleeps for 7 hours from 7:40 a.m. At what time does the baby wake up?

Step 1: $7:40 + 7 \text{ hours} = 14:40$.

Step 2: $14 - 12 = 2$, so the time is in the afternoon.

Final Answer: 2:40 p.m.

4(ii). Lusajo stays for 840 hours. How many days is that?

Step 1: $840 / 24 = 35$.

Final Answer: 35 days

4(iii). Yusufu arrives at 8:08 a.m., which is 30 minutes late. What was his intended arrival time?

Step 1: 8:08 minus 8 minutes = 8:00.

Step 2: Subtract the remaining 22 minutes to obtain 7:38.

Final Answer: 7:38 a.m.

5(i). Multiply 5 kg 50 g by 50.

Step 1: 5 kg 50 g = 5050 g.

Step 2: $5050 \times 50 = 252500$ g.

Step 3: 252500 g = 252 kg 500 g.

Final Answer: 252 kg 500 g

5(ii). Convert 8 $\frac{3}{4}$ L to ml.

Step 1: $8 \frac{3}{4} = 8.75$.

Step 2: $8.75 \times 1000 = 8750$.

Final Answer: 8750 ml

5(iii). Divide 607 km 50 m by 50.

Step 1: 607 km 50 m = 607050 m.

Step 2: $607050 / 50 = 12141$ m.

Final Answer: 12 km 141 m

6(i). Given $x = 2$, $y = 3$ and $2x + y - z = 12$, find z .

Step 1: $2 \times 2 + 3 = 7$.

Step 2: $7 - z = 12$, so $z = 7 - 12 = -5$.

Step 3: Check: $7 - (-5) = 12$.

Final Answer: $z = -5$

6(ii). Simplify $4x^2yz^3$ divided by $2xyz$.

Step 1: Coefficients: $4/2 = 2$; $x^2/x = x$; $y/y = 1$; $z^3/z = z^2$.

Step 2: Cancel the nonzero factors x , y and z .

Final Answer: $2xz^2$, with x , y and z nonzero.

6(iii). Twice a number minus 6 equals 12. Find the number.

Step 1: $2n = 12 + 6 = 18$.

Step 2: $18 / 2 = 9$.

Final Answer: 9

7(i). Find the mean of 45, 62, 38, 91 and 29.

Step 1: $45 + 62 + 38 + 91 + 29 = 265$.

Step 2: $265 / 5 = 53$.

Final Answer: 53

7(ii). The mean of 45, 90, 30, 29 and y is 56. Find y .

Step 1: Total = $56 \times 5 = 280$.

Step 2: Known total = $45 + 90 + 30 + 29 = 194$.

Step 3: $280 - 194 = 86$.

Final Answer: 86

8(i). A square has area 729 cm². Find its perimeter.

Step 1: $27 \times 27 = 729$, so the side is 27 cm.

Step 2: $4 \times 27 = 108$.

Final Answer: 108 cm

8(ii). Find the perimeter of the intended semicircle with diameter 28 cm.

Step 1: Assuming a semicircle and $\pi = 22/7$, the curved arc is $(22/7) \times 28 / 2 = 44$ cm.

Step 2: $44 + 28 = 72$.

Final Answer: 72 cm if the shape is a semicircle, using $\pi = 22/7$; the imprecise circular-segment drawing alone is insufficient.

8(iii). A circle has diameter 2.8 m. Find its area.

Step 1: Radius = $2.8 / 2 = 1.4$ m.

Step 2: Area = $\pi \times r^2$.

Step 3: Using $22/7$: $(22/7) \times 1.4 \times 1.4 = 6.16$.

Final Answer: 1.96 π m²; approximately 6.16 m² using $\pi = 22/7$.

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Source Checks

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